Time: 1 hr. Paper Code: CSC-501	B. Tech. (V Semester) 1st Sessional Test 2025-26 Object Oriented Programming	Max. Marks: 12

Note: Attempt All Questions. Make suitable assumptions wherever necessary.

Q1. **CO1**	(a) List any 5 key differences between procedure-oriented programming and object-oriented programming. (b) Explain the following terms with suitable code snippet: this pointer, Default Arguments, Function overloading, Virtual function	[2+2] BL-1,4
Q2. **CO2**	Develop a C++ program to efficiently manage and analyze IoT sensor data. Create two classes: TemperatureSensor, which stores a private temperature reading in °C, and HumiditySensor, which stores a private humidity reading in %. The program should prompt the user to enter the number of sensors, then collect temperature and humidity readings for each sensor. After gathering the data, display all individual sensor readings clearly. Finally, define a friend function named computeAverage() that directly accesses the private members of both classes to calculate and display the average temperature and average humidity.	[4] BL-6
Q3. **CO3**	Design a C++ program to model the staff hierarchy of CityCare Hospital. Staff is the base class with common attributes and a static member staff_count. Doctor, Nurse, and Technician inherit publicly from Staff as shown in figure. Each derived class adds role-specific attributes. Constructors increment staff_count; destructor decrements it and logs removal. Static showCount() displays current count. Create two objects for each derived class. One using the default constructor (giving some default values) and one using the parameterized constructor. Track and display the total number of active staff using the static counter before and after creating or deleting objects. Show the proper execution order of constructors and destructors. Display complete details of each staff member.	[4] BL-6

